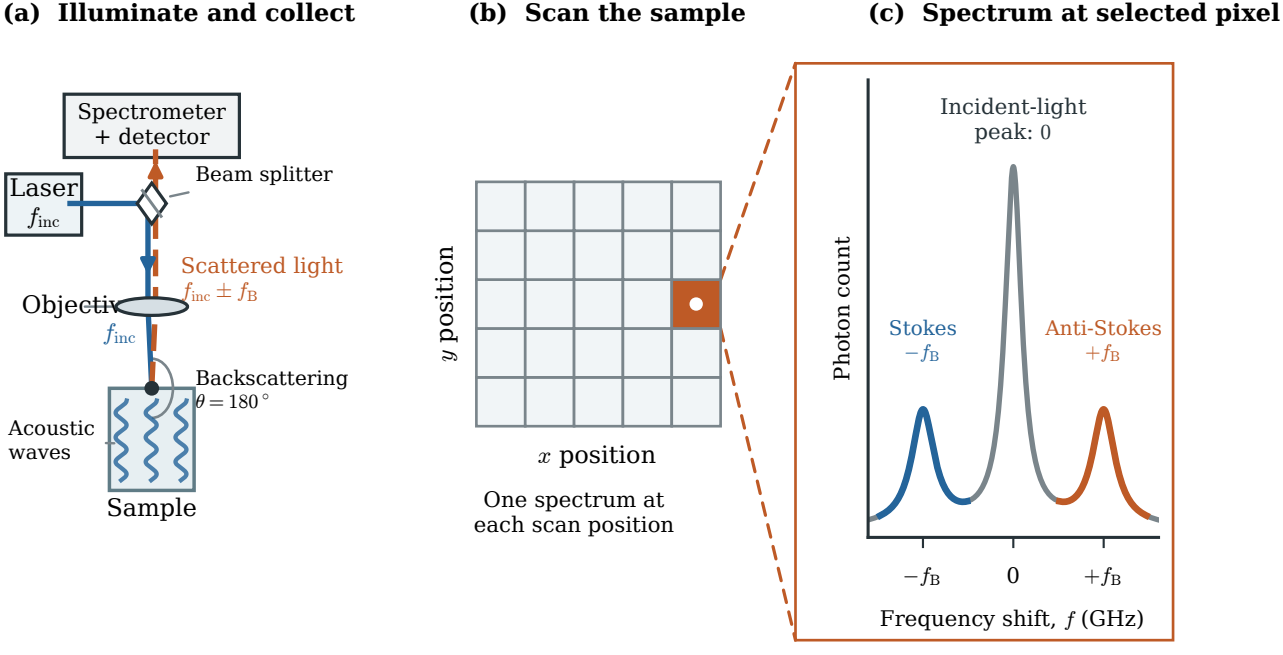
(a) Illuminate and collect
Spectrometer + detector
Laser
f_{inc}
Beam splitter
Scattered light
$f_{inc} \pm f_B$
Objectiv
f_{inc}
Backscattering
$\theta = 180°$
Acoustic waves
Sample
(b) Scan the sample
y position
x position
One spectrum at each scan position
(c) Spectrum at selected pixel
Incident-light peak: 0
Photon count
Stokes
$-f_B$
Anti-Stokes
$+f_B$
$-f_B$
0
$+f_B$
Frequency shift, f (GHz)